Solaris Analytics

ADS508 | Group 7 | Authors: Luigi Salemi, Francisco Hernandez

This notebook covers ingestion and EDA for all primary datasets:

#	Dataset	Source	Type
1	Houston Solar Irradiance 2000–2019 (NREL NSRDB)	S3	CSV — irradiance + weather
2	GPVS-Faults PV Sensor Readings	S3	CSV — electrical sensor + fault labels
3	Solar Panel Images (`pythonafroz`)	KaggleHub	Images — clean vs faulty
4	PV Panel Defect Dataset (`alicjalena`)	KaggleHub	Images — 6 defect classes

S3 Bucket: `s3://solaris-analytics-bucket/`
GitHub: https://github.com/darth-franky/Applied-Data-Science/tree/main/ADS508

In [34]:
```
import subprocess
subprocess.run(['pip', 'install', 'kagglehub', '-q'])

import os
import boto3
import sagemaker
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import seaborn as sns
import kagglehub
from pathlib import Path

session = sagemaker.Session()
s3 = boto3.client('s3')
role = sagemaker.get_execution_role()

BUCKET = 'solaris-analytics-bucket'
IMAGE_EXTENSIONS = {'.jpg', '.jpeg', '.png', '.bmp', '.tiff'}

print(f'SageMaker Role: {role}')
print(f'S3 Bucket: s3://{BUCKET}/')
```

```
SageMaker Role: arn:aws:iam::187806919439:role/LabRole
S3 Bucket: s3://solaris-analytics-bucket/
```

In [35]:
```
# List ALL objects in the bucket to find the exact key
response = s3.list_objects_v2(Bucket=BUCKET)
for obj in response.get('Contents', []):
    print(obj['Key'])
```

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CSV_Files/F0L.csv
CSV_Files/F0M.csv
CSV_Files/F1L.csv
CSV_Files/F1M.csv
CSV_Files/F2L.csv
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```

1. Ingest Houston Solar Irradiance Dataset (NREL NSRDB) from S3

File: `Houston_Solar_2000_2019.csv`

S3 path: `s3://solaris-analytics-bucket/Houston_Solar_2000_2019.csv`

Key fields: `Timestamp`, `GHI`, `DNI`, `DHI`, `Temperature`, `Wind_Speed`
Records: ~350,000 rows at 30-min intervals (2000-2019)

In [36]:
```
import os

# Create folder
os.makedirs('/home/sagemaker-user/dataset1', exist_ok=True)

# Set paths
nsrdb_s3_key = 'Houston_Solar_2000_2019.csv'
nsrdb_local  = '/home/sagemaker-user/dataset1/Houston_Solar_2000_2019.csv'

# Download from S3
print(f'Downloading s3://{BUCKET}/{nsrdb_s3_key} ...')
s3.download_file(BUCKET, nsrdb_s3_key, nsrdb_local)
print('Download complete.')

# Load into DataFrame
nsrdb_df = pd.read_csv(nsrdb_local, parse_dates=['Timestamp'])
print(f'Shape: {nsrdb_df.shape}')
nsrdb_df.head()
```

```
Downloading s3://solaris-analytics-bucket/Houston_Solar_2000_2019.csv ...
Download complete.
Shape: (350640, 6)
```

Out[36]:

	Timestamp	GHI	DNI	DHI	Temperature	Wind_Speed
0	2000-01-01 00:00:00	0	0	0	11.9	0.7
1	2000-01-01 00:30:00	0	0	0	11.5	0.7
2	2000-01-01 01:00:00	0	0	0	11.1	0.8
3	2000-01-01 01:30:00	0	0	0	10.6	0.9
4	2000-01-01 02:00:00	0	0	0	10.1	1.0

1.1 NSRDB — Basic Info & Missing Values

In [37]:
```
print('=== Data Types ===')
print(nsrdb_df.dtypes)

print('\n=== Missing Values ===')
missing = nsrdb_df.isnull().sum()
print(missing[missing > 0] if missing.sum() > 0 else 'No missing values four

print('\n=== Descriptive Statistics ===')
display(nsrdb_df.describe().T)
```

```
=== Data Types ===
Timestamp      datetime64[ns]
GHI                     int64
DNI                     int64
DHI                     int64
Temperature           float64
Wind_Speed            float64
dtype: object

=== Missing Values ===
No missing values found.

=== Descriptive Statistics ===
```

	count	mean	min	25%	50%	75%	
Timestamp	350640	2009-12-31 11:45:00.000001024	2000-01-01 00:00:00	2004-12-31 05:52:30	2009-12-31 11:45:00	2014-12-31 17:37:30	2 1 23:3
GHI	350640.0	201.67449	0.0	0.0	0.0	365.0	10
DNI	350640.0	197.567154	0.0	0.0	0.0	348.0	10
DHI	350640.0	79.42834	0.0	0.0	0.0	120.0	6
Temperature	350640.0	20.82322	-7.0	15.2	22.2	26.6	
Wind_Speed	350640.0	2.218328	0.2	1.3	2.0	2.9	

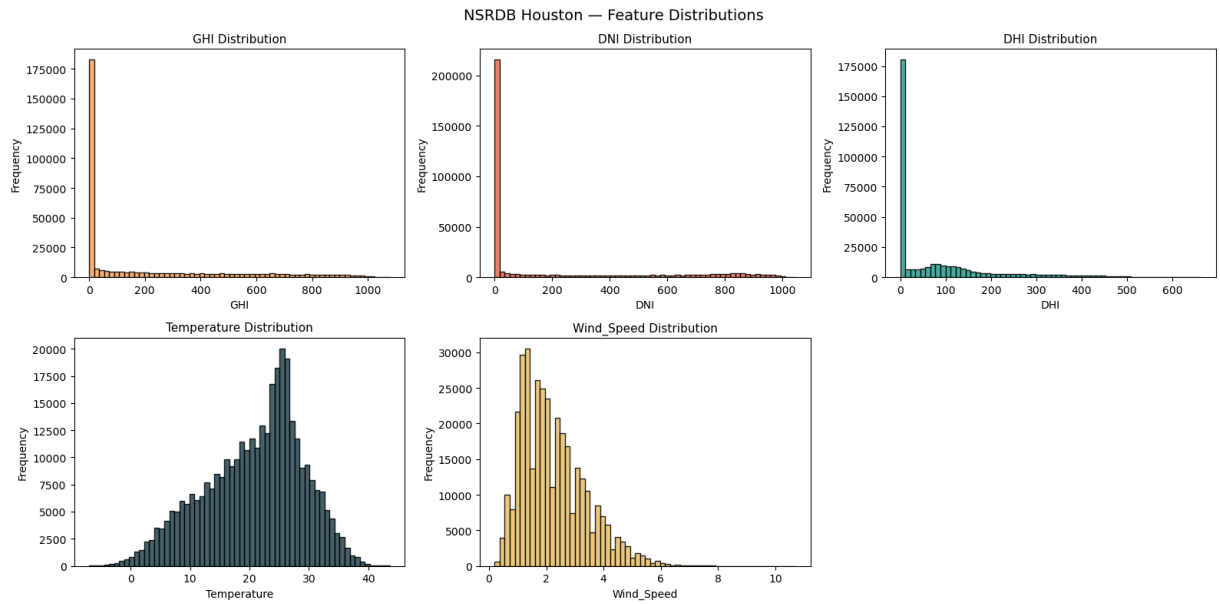
1.2 NSRDB — Irradiance & Weather Distributions

In [38]:
```
irradiance_cols = ['GHI', 'DNI', 'DHI']
weather_cols    = ['Temperature', 'Wind_Speed']
all_numeric     = irradiance_cols + weather_cols

fig, axes = plt.subplots(2, 3, figsize=(16, 8))
axes = axes.flatten()
colors = ['#F4A261', '#E76F51', '#2A9D8F', '#264653', '#E9C46A']

for i, col in enumerate(all_numeric):
    axes[i].hist(nsrdb_df[col].dropna(), bins=60, color=colors[i], edgecolor
    axes[i].set_title(f'{col} Distribution', fontsize=11)
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Frequency')

axes[5].set_visible(False)
plt.suptitle('NSRDB Houston — Feature Distributions', fontsize=14)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_distributions.png', dpi=150)
plt.show()
print('Note: High frequency of GHI=0 reflects nighttime — these rows may be
```




```
Note: High frequency of GHI=0 reflects nighttime — these rows may be filtere
d before forecasting.
```

1.3 NSRDB — Seasonal & Temporal Bias Check

In [39]:
```
nsrdb_df['Year']  = nsrdb_df['Timestamp'].dt.year
nsrdb_df['Month'] = nsrdb_df['Timestamp'].dt.month
nsrdb_df['Hour']  = nsrdb_df['Timestamp'].dt.hour

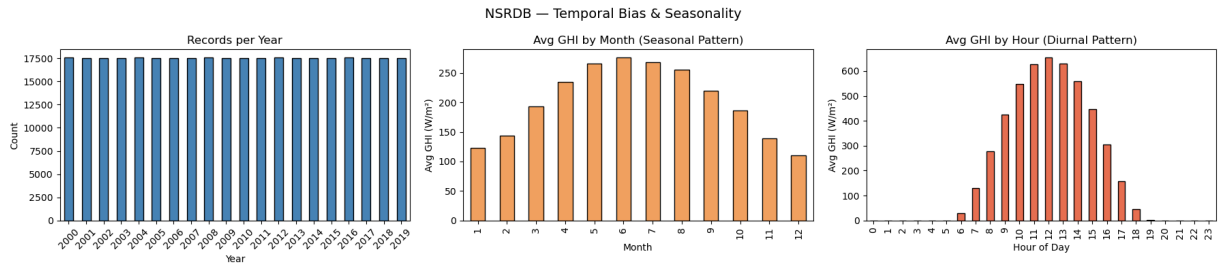
fig, axes = plt.subplots(1, 3, figsize=(18, 4))

nsrdb_df['Year'].value_counts().sort_index().plot(
    kind='bar', ax=axes[0], color='steelblue', edgecolor='black')
axes[0].set_title('Records per Year')
axes[0].set_xlabel('Year')
axes[0].set_ylabel('Count')
axes[0].tick_params(axis='x', rotation=45)

nsrdb_df.groupby('Month')['GHI'].mean().plot(
    kind='bar', ax=axes[1], color='#F4A261', edgecolor='black')
axes[1].set_title('Avg GHI by Month (Seasonal Pattern)')
axes[1].set_xlabel('Month')
axes[1].set_ylabel('Avg GHI (W/m²)')

nsrdb_df.groupby('Hour')['GHI'].mean().plot(
    kind='bar', ax=axes[2], color='#E76F51', edgecolor='black')
axes[2].set_title('Avg GHI by Hour (Diurnal Pattern)')
axes[2].set_xlabel('Hour of Day')
axes[2].set_ylabel('Avg GHI (W/m²)')

plt.suptitle('NSRDB — Temporal Bias & Seasonality', fontsize=14)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_temporal_bias.png', dpi=150)
plt.show()
print('Note: NSRDB is U.S.-centric (Houston only). Seasonal and diurnal patt
```




```
Note: NSRDB is U.S.-centric (Houston only). Seasonal and diurnal patterns ar
e expected physical behavior.
```

1.4 NSRDB — GHI Irradiance Category Distribution

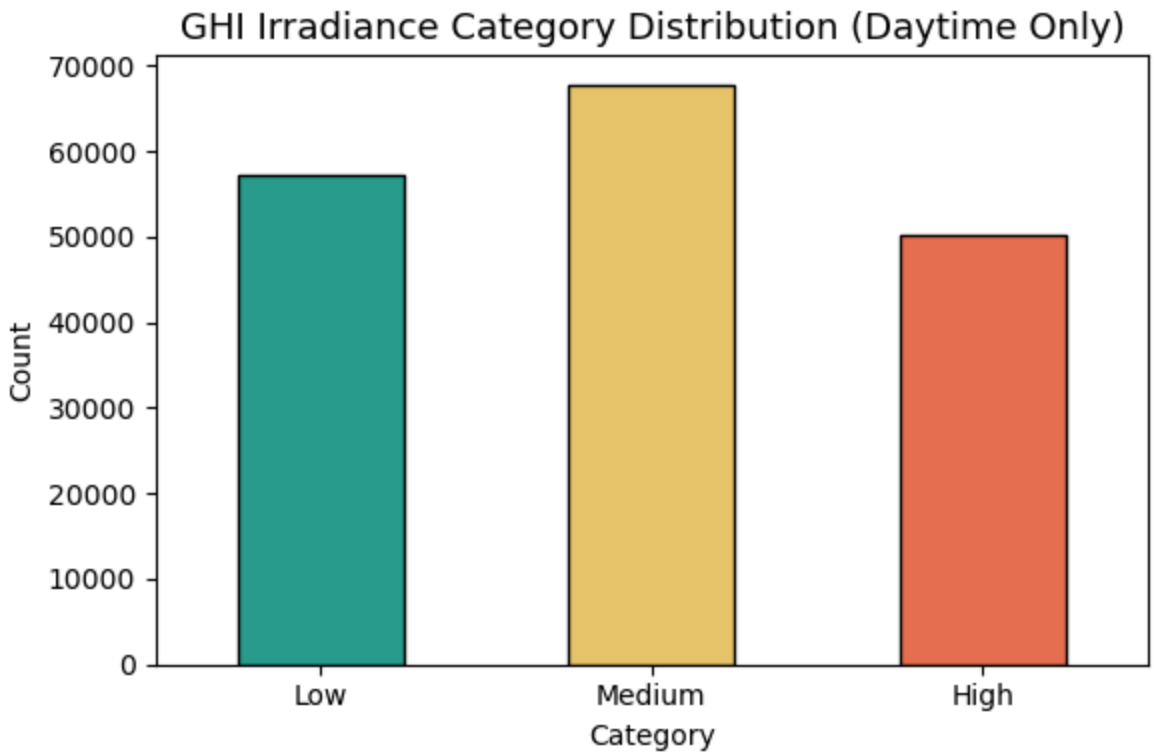
Per the design doc, GHI will be bucketed into **Low / Medium / High** for classification tasks.

In [40]:
```
daytime_df = nsrdb_df[nsrdb_df['GHI'] > 0].copy()
daytime_df['GHI_Category'] = pd.cut(
    daytime_df['GHI'],
    bins=[0, 200, 600, daytime_df['GHI'].max() + 1],
    labels=['Low', 'Medium', 'High']
)

print('=== GHI Category Distribution (daytime only) ===')
print(daytime_df['GHI_Category'].value_counts())

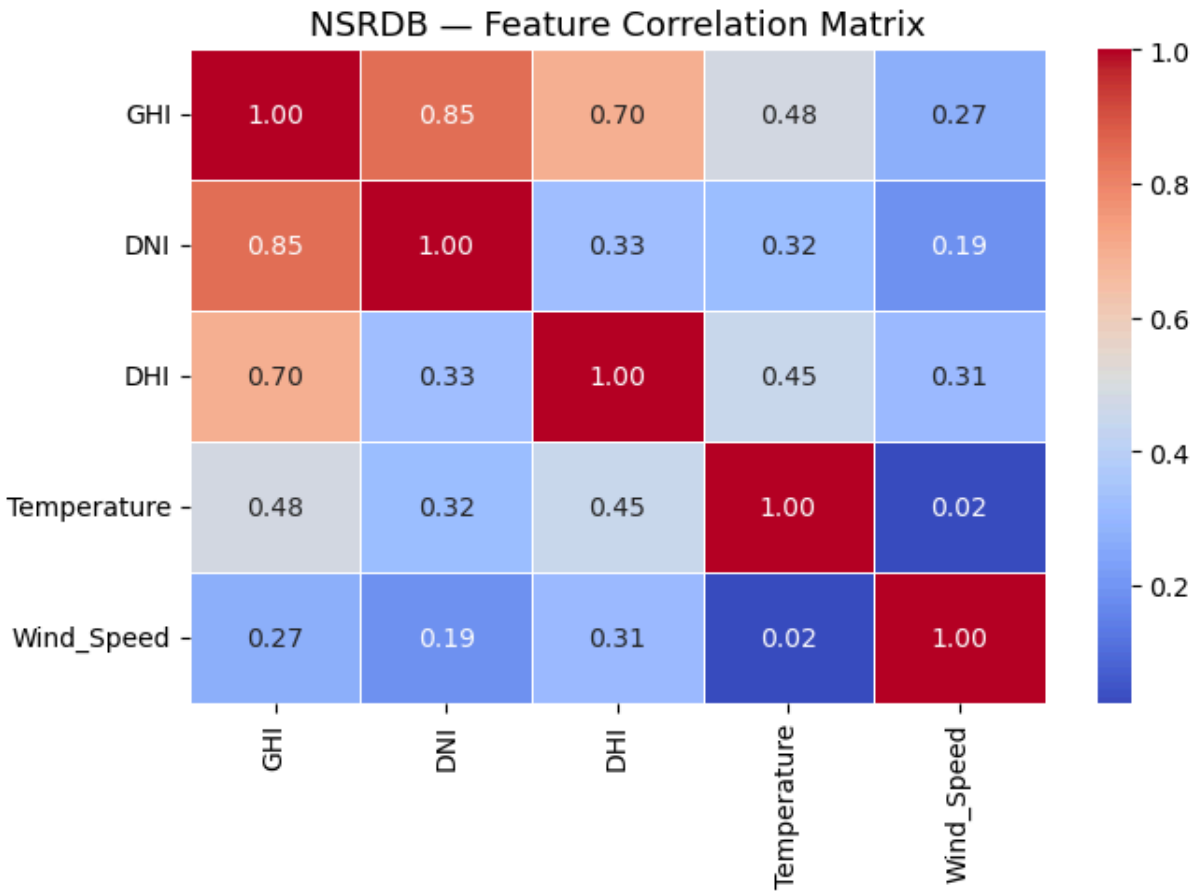
fig, ax = plt.subplots(figsize=(6, 4))
daytime_df['GHI_Category'].value_counts().reindex(['Low','Medium','High']).p
    kind='bar', ax=ax, color=['#2A9D8F','#E9C46A','#E76F51'], edgecolor='bla
)
ax.set_title('GHI Irradiance Category Distribution (Daytime Only)', fontsize
ax.set_xlabel('Category')
ax.set_ylabel('Count')
ax.tick_params(axis='x', rotation=0)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_ghi_categories.png', dpi=150)
plt.show()
```

```
=== GHI Category Distribution (daytime only) ===
GHI_Category
Medium    67793
Low       57143
High      50195
Name: count, dtype: int64
```



1.5 NSRDB — Correlation Heatmap

In [41]:
```
fig, ax = plt.subplots(figsize=(7, 5))
sns.heatmap(nsrdb_df[all_numeric].corr(), annot=True, fmt='.2f', cmap='coolw
ax.set_title('NSRDB — Feature Correlation Matrix', fontsize=13)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_correlation.png', dpi=150)
plt.show()
```



2. Ingest GPVS-Faults Sensor Dataset from S3

S3 path: `s3://solaris-analytics-bucket/dataset2/gpvs-faults/`
Key fields: voltage, current, power, fault_label
Records: 10,000+

In [42]:
```
response  = s3.list_objects_v2(Bucket=BUCKET)
gpvs_keys = [obj['Key'] for obj in response.get('Contents', [])]

local_gpvs_dir = '/home/sagemaker-user/dataset2/gpvs-faults/'
os.makedirs(local_gpvs_dir, exist_ok=True)

for key in gpvs_keys:
    filename = key.split('/')[-1]
    if filename:
        local_path = os.path.join(local_gpvs_dir, filename)
        s3.download_file(BUCKET, key, local_path)
        print(f'Downloaded: {key}')

csv_files = list(Path(local_gpvs_dir).glob('*.csv')) + list(Path(local_gpvs_
dfs = []
for f in csv_files:
    sep = '\t' if f.suffix == '.tsv' else ','
    df_temp = pd.read_csv(f, sep=sep)
    df_temp['source_file'] = f.name
    dfs.append(df_temp)
```

```
    print(f'Loaded {f.name}: {df_temp.shape}')

gpvs_df = pd.concat(dfs, ignore_index=True)
print(f'\nCombined GPVS shape: {gpvs_df.shape}')
gpvs_df.head()
```

```
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Downloaded: CSV_Files/F0M.csv
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Downloaded: CSV_Files/F2M.csv
Downloaded: CSV_Files/F3L.csv
Downloaded: CSV_Files/F3M.csv
Downloaded: CSV_Files/F4L.csv
Downloaded: CSV_Files/F4M.csv
Downloaded: CSV_Files/F5L.csv
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Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (45).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (46).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (47).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (48).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (5).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (50).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (51).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (52).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (54).jpg
```

```
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (55).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (56).jpg
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Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (69).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (7).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (70).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (71).jpg
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Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (98).jpg
Downloaded: panel-images/pv-defect/train/Physical-Damage/Physical (99).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (1).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (10).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (100).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (101).jpg
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Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (107).jpg
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Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (109).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (11).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (111).jpg
```

```
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (113).jpg
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Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (115).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (116).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (119).jpg
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```

```
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (174).jpg
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Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (176).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (18).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (19).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (2).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (20).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (21).jpg
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Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (30).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (31).jpg
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Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (37).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (38).jpg
Downloaded: panel-images/pv-defect/train/Snow-Covered/Snow (39).jpg
Loaded F0L.csv: (143715, 15)
Loaded F0M.csv: (141014, 15)
Loaded F1L.csv: (129013, 15)
Loaded F1M.csv: (139014, 15)
Loaded F2L.csv: (142128, 15)
Loaded F2M.csv: (144015, 15)
Loaded F3L.csv: (103497, 15)
Loaded F3M.csv: (69967, 15)
Loaded F4L.csv: (144014, 15)
Loaded F4M.csv: (144014, 15)
Loaded F5L.csv: (143015, 15)
Loaded F5M.csv: (144014, 15)
Loaded F6L.csv: (144015, 15)
Loaded F6M.csv: (144015, 15)
Loaded F7L.csv: (144015, 15)
Loaded F7M.csv: (144015, 15)
Loaded Houston_Solar_2000_2019.csv: (350640, 7)

Combined GPVS shape: (2514120, 21)
```

Out[42]:

	Time	Ipv	Vpv	Vdc	ia	ib	ic	
0	0.000028	1.572327	101.348877	144.140625	-0.135133	0.490112	-0.354985	41.7
1	0.000128	1.503265	101.458740	143.554688	-0.108277	0.510254	-0.388555	46.8
2	0.000228	1.492859	101.574707	143.554688	-0.168702	0.496826	-0.334844	51.0
3	0.000328	1.558136	101.312256	143.261719	-0.135133	0.510254	-0.361699	55.8
4	0.000428	1.631927	101.141357	143.847656	-0.202271	0.503540	-0.321416	60.0

5 rows × 21 columns

2.1 GPVS — Basic Info, Missing Values & Zero-Value Check

In [43]:

```
print('=== Data Types ===')
print(gpvs_df.dtypes)

print('\n=== Missing Values ===')
missing = gpvs_df.isnull().sum()
print(missing[missing > 0] if missing.sum() > 0 else 'No missing values four

numeric_cols = gpvs_df.select_dtypes(include=[np.number]).columns.tolist()

print('\n=== Zero-Value Counts (potential sensor outage indicator) ===')
zero_counts = (gpvs_df[numeric_cols] == 0).sum()
zero_pct    = (zero_counts / len(gpvs_df) * 100).round(2)
print(pd.DataFrame({'Zero Count': zero_counts, 'Zero %': zero_pct}).to_strir
print('\nNote: High zero % in voltage/current/power may indicate sensor outa
```

```
=== Data Types ===
Time           float64
Ipv            float64
Vpv            float64
Vdc            float64
ia             float64
ib             float64
ic             float64
va             float64
vb             float64
vc             float64
Iabc           float64
If             float64
Vabc           float64
Vf             float64
source_file     object
Timestamp       object
GHI            float64
DNI            float64
DHI            float64
Temperature    float64
Wind_Speed     float64
dtype: object

=== Missing Values ===
Time            350640
Ipv             350640
Vpv             350640
Vdc             350640
ia              350640
ib              350640
ic              350640
va              350640
vb              350640
vc              350640
Iabc            350640
If              350640
Vabc            350640
Vf              350640
Timestamp      2163480
GHI            2163480
DNI            2163480
DHI            2163480
Temperature    2163480
Wind_Speed     2163480
dtype: int64

=== Zero-Value Counts (potential sensor outage indicator) ===
             Zero Count  Zero %
Time                  0    0.00
Ipv                  48    0.00
Vpv                   0    0.00
Vdc                   1    0.00
ia                    0    0.00
ib                21441    0.85
ic                    0    0.00
```

```
va                   55    0.00
vb                   55    0.00
vc                    0    0.00
Iabc                  0    0.00
If                    0    0.00
Vabc                  0    0.00
Vf                    0    0.00
GHI              175509    6.98
DNI              195739    7.79
DHI              175509    6.98
Temperature          90    0.00
Wind_Speed            0    0.00

Note: High zero % in voltage/current/power may indicate sensor outages needi
ng removal in data prep.
```

2.2 GPVS — Feature Distributions & Correlation

In [44]:

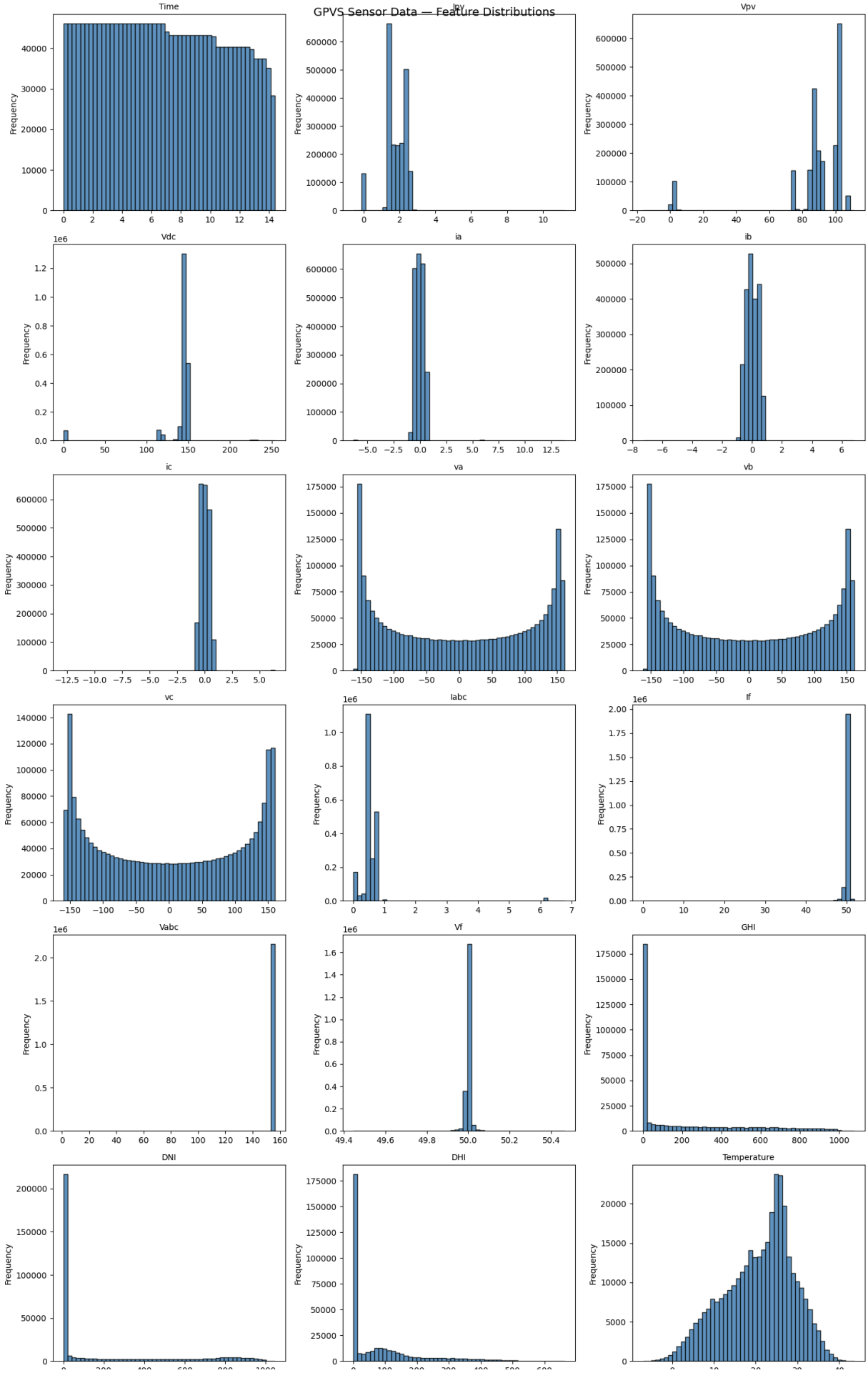
```
plot_cols = [c for c in numeric_cols if c not in (fault_col_candidates or []
n = len(plot_cols)
cols_per_row = 3
rows = (n + cols_per_row - 1) // cols_per_row

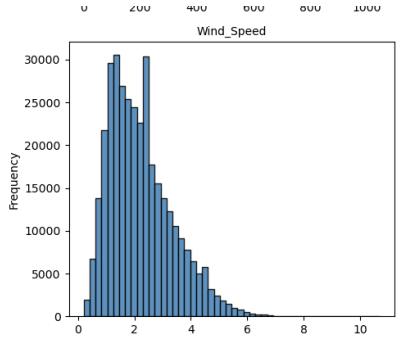
fig, axes = plt.subplots(rows, cols_per_row, figsize=(15, rows * 4))
axes = axes.flatten()

for i, col in enumerate(plot_cols):
    axes[i].hist(gpvs_df[col].dropna(), bins=50, color='steelblue', edgecolo
    axes[i].set_title(col, fontsize=10)
    axes[i].set_ylabel('Frequency')

for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)

plt.suptitle('GPVS Sensor Data — Feature Distributions', fontsize=14)
plt.tight_layout()
plt.savefig('/tmp/gpvs_distributions.png', dpi=150)
plt.show()
```


GPVS Sensor Data — Feature Distributions
Time
Ipv
Vpv
Vdc
ia
ib
ic
va
vb
vc
Iabc
If
Vabc
Vf
GHI
DNI
DHI
Temperature
Frequency



---

3. Ingest Image Datasets via KaggleHub → Upload to S3

Two image datasets downloaded from Kaggle and persisted to S3:

- `pythonafroz/solar-panel-images` → binary: clean vs faulty panels
- `alicjalena/pv-panel-defect-dataset` → multi-class: 6 defect types

In [45]:
```
def upload_dir_to_s3(local_dir, s3_prefix, bucket):
    """Recursively upload a local directory to an S3 prefix."""
    count = 0
    for root, _, files in os.walk(local_dir):
        for file in files:
            local_path = os.path.join(root, file)
            rel_path   = os.path.relpath(local_path, local_dir)
            s3_key     = s3_prefix + rel_path.replace(os.sep, '/')
            s3.upload_file(local_path, bucket, s3_key)
            count += 1
    return count

def count_images_by_class(base_path):
    """Return dict of {class_name: image_count} from a folder of class subfo
    counts = {}
    for item in Path(base_path).iterdir():
        if item.is_dir():
            n = sum(1 for f in item.rglob('*') if f.suffix.lower() in IMAGE_
            if n > 0:
                counts[item.name] = n
    return counts

print('Helper functions defined.')
```

```
Helper functions defined.
```

In [46]:
```
# Download Solar Panel Images (pythonafroz)
print('Downloading pythonafroz/solar-panel-images ...')
solar_panel_path = kagglehub.dataset_download('pythonafroz/solar-panel-image
print(f'Downloaded to: {solar_panel_path}')

n = upload_dir_to_s3(solar_panel_path, 'panel-images/solar-panel-images/', E
print(f'Uploaded {n} files to s3://{BUCKET}/panel-images/solar-panel-images/
```

```
Downloading pythonafroz/solar-panel-images ...
Downloaded to: /home/sagemaker-user/.cache/kagglehub/datasets/pythonafroz/so
lar-panel-images/versions/2
Uploaded 891 files to s3://solaris-analytics-bucket/panel-images/solar-panel
-images/
```

In [47]:
```
# Download PV Panel Defect Dataset (alicjalena)
print('Downloading alicjalena/pv-panel-defect-dataset ...')
pv_defect_path = kagglehub.dataset_download('alicjalena/pv-panel-defect-data
print(f'Downloaded to: {pv_defect_path}')

n = upload_dir_to_s3(pv_defect_path, 'panel-images/pv-defect/', BUCKET)
print(f'Uploaded {n} files to s3://{BUCKET}/panel-images/pv-defect/')
```

```
Downloading alicjalena/pv-panel-defect-dataset ...
Downloaded to: /home/sagemaker-user/.cache/kagglehub/datasets/alicjalena/pv-
panel-defect-dataset/versions/1
Uploaded 1574 files to s3://solaris-analytics-bucket/panel-images/pv-defect/
```

3.1 Solar Panel Images (pythonafroz) — Distribution & Samples

In [48]:
```
solar_counts = count_images_by_class(solar_panel_path)
print('=== Solar Panel Images — Images per Class ===')
for cls, cnt in sorted(solar_counts.items()):
    print(f'  {cls}: {cnt} images')
print(f'Total: {sum(solar_counts.values())}')

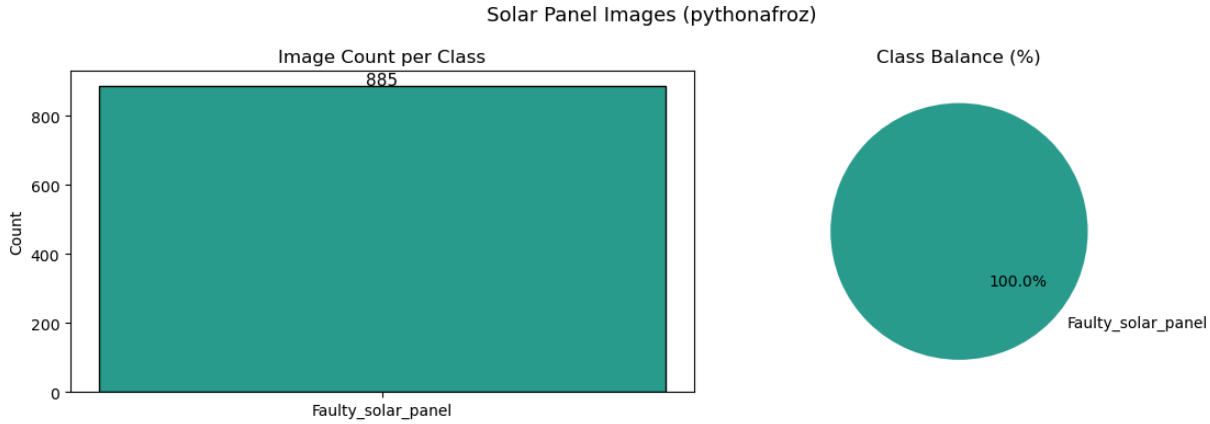
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].bar(solar_counts.keys(), solar_counts.values(), color=['#2A9D8F','#E
for i, (cls, cnt) in enumerate(solar_counts.items()):
    axes[0].text(i, cnt + 5, str(cnt), ha='center', fontsize=11)
axes[0].set_title('Image Count per Class', fontsize=12)
axes[0].set_ylabel('Count')

axes[1].pie(solar_counts.values(), labels=solar_counts.keys(),
            autopct='%1.1f%%', colors=['#2A9D8F','#E76F51'], startangle=140)
axes[1].set_title('Class Balance (%)', fontsize=12)

plt.suptitle('Solar Panel Images (pythonafroz)', fontsize=13)
plt.tight_layout()
plt.savefig('/tmp/solar_panel_distribution.png', dpi=150)
plt.show()
```

```
=== Solar Panel Images — Images per Class ===
  Faulty_solar_panel: 885 images
Total: 885
```



In [49]:
```
# Sample one image per class
classes = sorted(solar_counts.keys())
fig, axes = plt.subplots(1, len(classes), figsize=(6 * len(classes), 5))
if len(classes) == 1:
    axes = [axes]

for ax, cls in zip(axes, classes):
    class_dir = Path(solar_panel_path) / cls
    imgs = [f for f in class_dir.rglob('*') if f.suffix.lower() in IMAGE_EXT
    if imgs:
        ax.imshow(mpimg.imread(str(imgs[0])))
        ax.set_title(cls, fontsize=12)
        ax.axis('off')

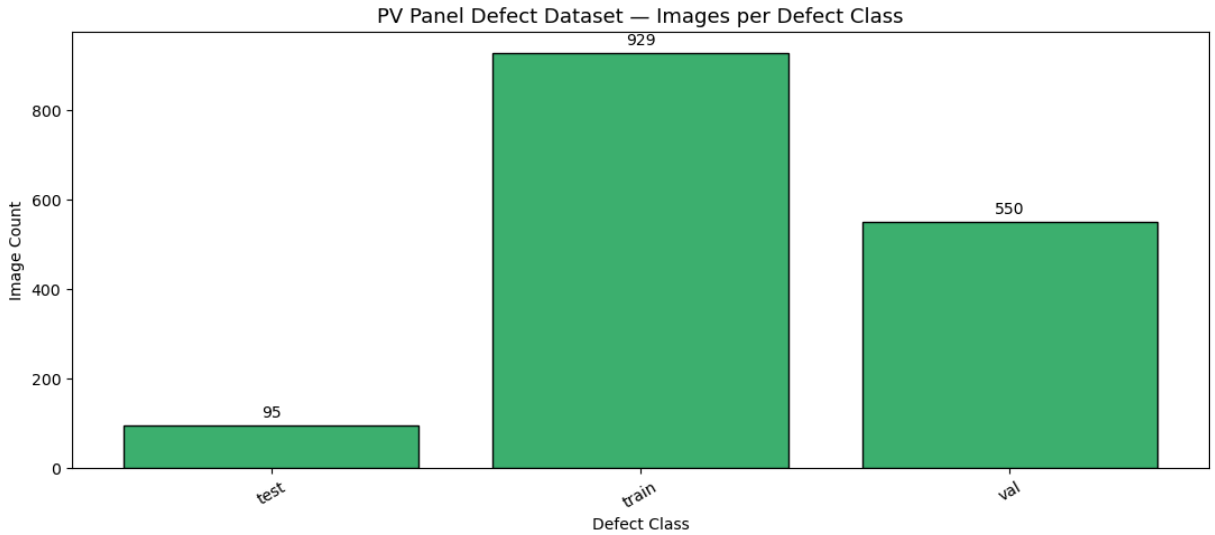
plt.suptitle('Solar Panel Images — Sample per Class', fontsize=13)
plt.tight_layout()
plt.savefig('/tmp/solar_panel_samples.png', dpi=150)
plt.show()
```

3.2 PV Panel Defect Dataset (alicjalena) — Distribution & Samples

In [50]:
```
defect_counts = count_images_by_class(pv_defect_path)
print('=== PV Panel Defect Dataset — Images per Class ===')
for cls, cnt in sorted(defect_counts.items()):
    print(f'  {cls}: {cnt} images')
print(f'Total: {sum(defect_counts.values())}')

fig, ax = plt.subplots(figsize=(11, 5))
bars = ax.bar(defect_counts.keys(), defect_counts.values(), color='mediumsea
ax.bar_label(bars, padding=3)
ax.set_title('PV Panel Defect Dataset — Images per Defect Class', fontsize=1
ax.set_xlabel('Defect Class')
ax.set_ylabel('Image Count')
ax.tick_params(axis='x', rotation=30)
plt.tight_layout()
plt.savefig('/tmp/pv_defect_distribution.png', dpi=150)
plt.show()
print('Note: Underrepresented defect classes will require augmentation or cl
```

```
=== PV Panel Defect Dataset — Images per Class ===
  test: 95 images
  train: 929 images
  val: 550 images
Total: 1574
```

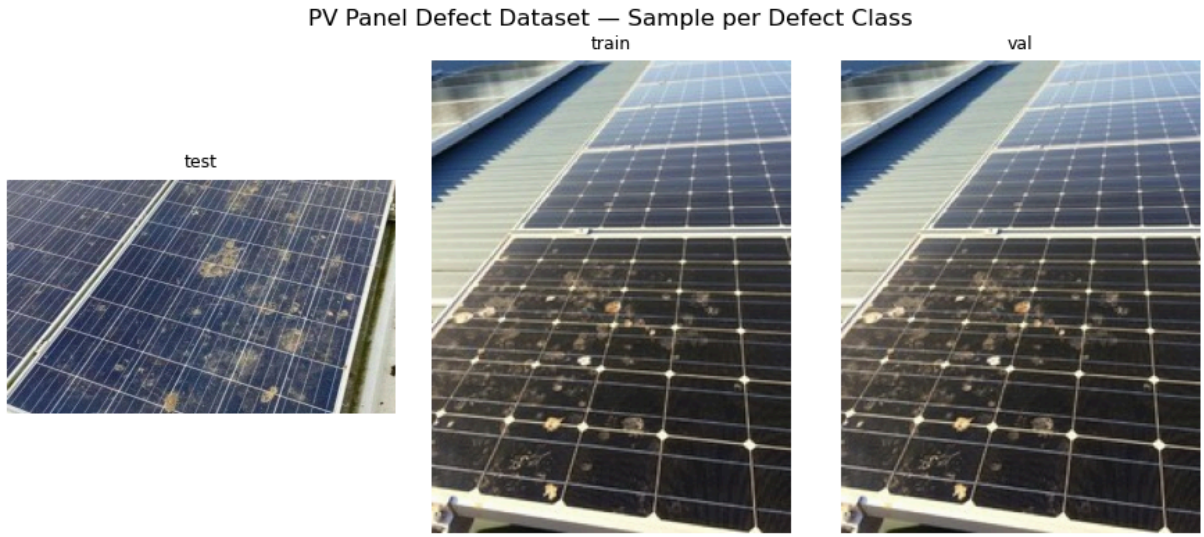



```
Note: Underrepresented defect classes will require augmentation or class wei
ghting during CNN training.
```

In [51]:
```
display_classes = sorted(defect_counts.keys())[:6]
fig, axes = plt.subplots(1, len(display_classes), figsize=(3 * len(display_c
if len(display_classes) == 1:
    axes = [axes]

for ax, cls in zip(axes, display_classes):
    class_dir = Path(pv_defect_path) / cls
    imgs = [f for f in class_dir.rglob('*') if f.suffix.lower() in IMAGE_EXT
    if imgs:
        ax.imshow(mpimg.imread(str(imgs[0])))
        ax.set_title(cls, fontsize=9)
        ax.axis('off')

plt.suptitle('PV Panel Defect Dataset — Sample per Defect Class', fontsize=1
plt.tight_layout()
plt.savefig('/tmp/pv_defect_samples.png', dpi=150)
plt.show()
```



---

4. Full EDA Summary

In [52]:
```
print('=' * 62)
print('       SOLARIS ANALYTICS — COMPLETE EDA SUMMARY')
print('=' * 62)

print(f'\n[1] NSRDB Houston Solar Irradiance (2000-2019)')
print(f'    Rows:       {len(nsrdb_df):,}')
print(f'    Columns:    {nsrdb_df.shape[1]}')
print(f'    Date Range: {nsrdb_df["Timestamp"].min()} to {nsrdb_df["Timestam
print(f'    Missing:    {nsrdb_df.isnull().sum().sum()}')
print(f'    GHI Range:  {nsrdb_df["GHI"].min()} - {nsrdb_df["GHI"].max()} W/
print(f'    Key Risk:   U.S.-centric (Houston only); seasonal/diurnal bias e

print(f'\n[2] GPVS-Faults Sensor Data')
print(f'    Rows:       {len(gpvs_df):,}')
print(f'    Columns:    {gpvs_df.shape[1]}')
print(f'    Missing:    {gpvs_df.isnull().sum().sum()}')
print(f'    Key Risk:   Single PV system source; check fault class imbalance

print(f'\n[3] Solar Panel Images (pythonafroz) — Binary Classification')
for cls, cnt in sorted(solar_counts.items()):
    print(f'    {cls}: {cnt} images')
print(f'    Total: {sum(solar_counts.values())} | Key Risk: Binary class bal

print(f'\n[4] PV Panel Defect Dataset (alicjalena) — Multi-class (6 classes)
for cls, cnt in sorted(defect_counts.items()):
    print(f'    {cls}: {cnt} images')
print(f'    Total: {sum(defect_counts.values())} | Key Risk: Multi-class imb

print(f'\nAll data stored at: s3://{BUCKET}/')
print(f'  nsrdb/                          <- NREL irradiance CSV')
print(f'  gpvs-faults/                    <- PV sensor fault CSV')
print(f'  panel-images/solar-panel-images/ <- pythonafroz images')
print(f'  panel-images/pv-defect/          <- alicjalena images')
```

```
print(f'\nGitHub: https://github.com/darth-franky/Applied-Data-Science/tree/
print('=' * 62)
```

```
==============================================================
       SOLARIS ANALYTICS — COMPLETE EDA SUMMARY
==============================================================

[1] NSRDB Houston Solar Irradiance (2000-2019)
    Rows:       350,640
    Columns:    9
    Date Range: 2000-01-01 00:00:00 to 2019-12-31 23:30:00
    Missing:    0
    GHI Range:  0 - 1078 W/m2
    Key Risk:   U.S.-centric (Houston only); seasonal/diurnal bias expected

[2] GPVS-Faults Sensor Data
    Rows:       2,514,120
    Columns:    21
    Missing:    17889840
    Key Risk:   Single PV system source; check fault class imbalance

[3] Solar Panel Images (pythonafroz) — Binary Classification
    Faulty_solar_panel: 885 images
    Total: 885 | Key Risk: Binary class balance

[4] PV Panel Defect Dataset (alicjalena) — Multi-class (6 classes)
    test: 95 images
    train: 929 images
    val: 550 images
    Total: 1574 | Key Risk: Multi-class imbalance

All data stored at: s3://solaris-analytics-bucket/
  nsrdb/                          <- NREL irradiance CSV
  gpvs-faults/                    <- PV sensor fault CSV
  panel-images/solar-panel-images/ <- pythonafroz images
  panel-images/pv-defect/          <- alicjalena images

GitHub: https://github.com/darth-franky/Applied-Data-Science/tree/main/ADS50
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==============================================================
```